place, it proactively identifies and addresses these compatibility problems earlier in the development cycle. Consequently, the integration test only encounters problems not caught by contract testing, streamlining the debugging process and enabling a faster feedback loop. By detecting these problems earlier, organizations can prevent them from propagating to integrated environments.

If your team already has a substantial number of integration tests, many of these tests can also be shifted earlier in the development process to the contract level. However, this doesn't entirely replace the need for integration or E2E testing because contract testing doesn't verify the following:

- Environment problems caused by configuration problems
- Network problems such as timeouts or dropped requests
- Data integrity relating to how the data is stored
- The end-user workflow journey and any consequences to other parts of the system
- Misunderstandings in the business requirements

If you do want to move some of your integration or E2E tests with contract tests, a good rule of thumb is to make sure that you're only moving the tests that can be duplicated. If the test can be covered already on the contract level, this gives you increased confidence that you don't need to have the same test on the integration or E2E level. We'll explore how you can move integration and E2E tests in more detail as part of chapter 12. Specifically, section 12.3 looks at converting an integration test to a contract test, while section 12.4 looks at converting an E2E test to a contract test.

2.3 Benefits of contract testing

In chapter 1, we explained a few benefits of contract testing. However, if you're still not convinced to consider contract testing, here are additional benefits that contract testing can provide to you and your teams:

- Cost saving benefits